

Four-Dimensional Mandelbrot Sets for Binary Self-Similar Sets of the Plane

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Parameters. $p = (c_0, c_1) \in \mathcal{P} := \{(c_0, c_1) \in \mathbb{C}^2 : |c_0|, |c_1| > 1\}$.

$$f_0(z) = -1 + z/c_0$$

$$f_1(z) = 1 + z/c_1$$

$$g_0(z) = -1 + \bar{z}/c_0$$

$$g_1(z) = 1 + \bar{z}/c_1$$

For a pair (h_0, h_1) below, let $K = K(p) \subset \mathbb{C}$ be the unique nonempty compact set with $K = h_0(K) \cup h_1(K)$.

Its connectedness locus is $\{p \in \mathcal{P} : K(p) \text{ is connected}\}$. Each column has its own K ; write $\mathcal{M} = \mathcal{M}^{\mathfrak{g}}$.

Direct (f_0, f_1)

$$p \in \mathcal{M} \iff 2 \in K/c_0 - K/c_1$$

Mixed (f_0, g_1)

$$p \in \mathcal{M}^{\mathfrak{g}} \iff 2 \in K/c_0 - \bar{K}/c_1$$

Reversing (g_0, g_1)

$$p \in \mathcal{M}^{\mathfrak{g}} \iff 2 \in \bar{K}/c_0 - K/c_1$$

Set notation: $K/c = \{z/c : z \in K\}$, $\bar{K} = \{\bar{z} : z \in K\}$, $A - B = \{a - b : a \in A, b \in B\}$.

Main theorem [4]. $\mathcal{M}$ is path connected and regular closed: $\mathcal{M} = \overline{\text{int } \mathcal{M}}$. Interior and closure in $\mathcal{P}$.

All three loci. Every real connected parameter is a limit of interior parameters in $\mathcal{P}$.

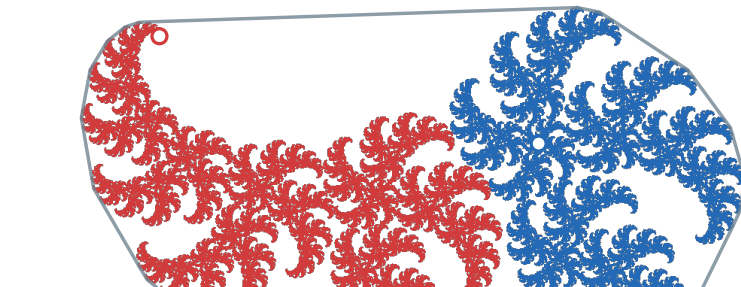


Explore the atlas

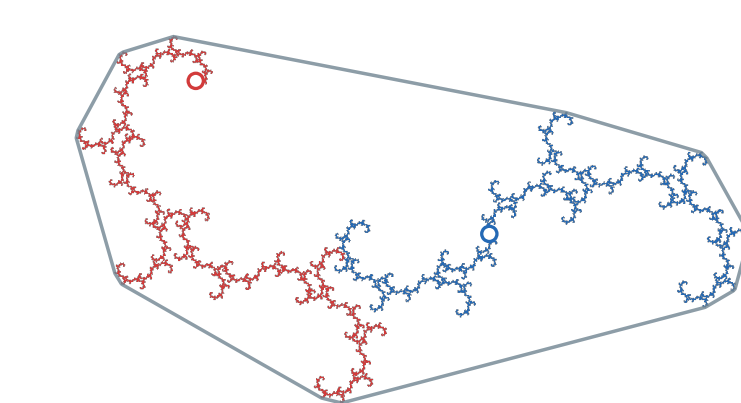
1 Equal-contraction atlas of $\mathcal{M}$

A 3D slice: $|c_0| = |c_1|$. Two rotations and one common scale.

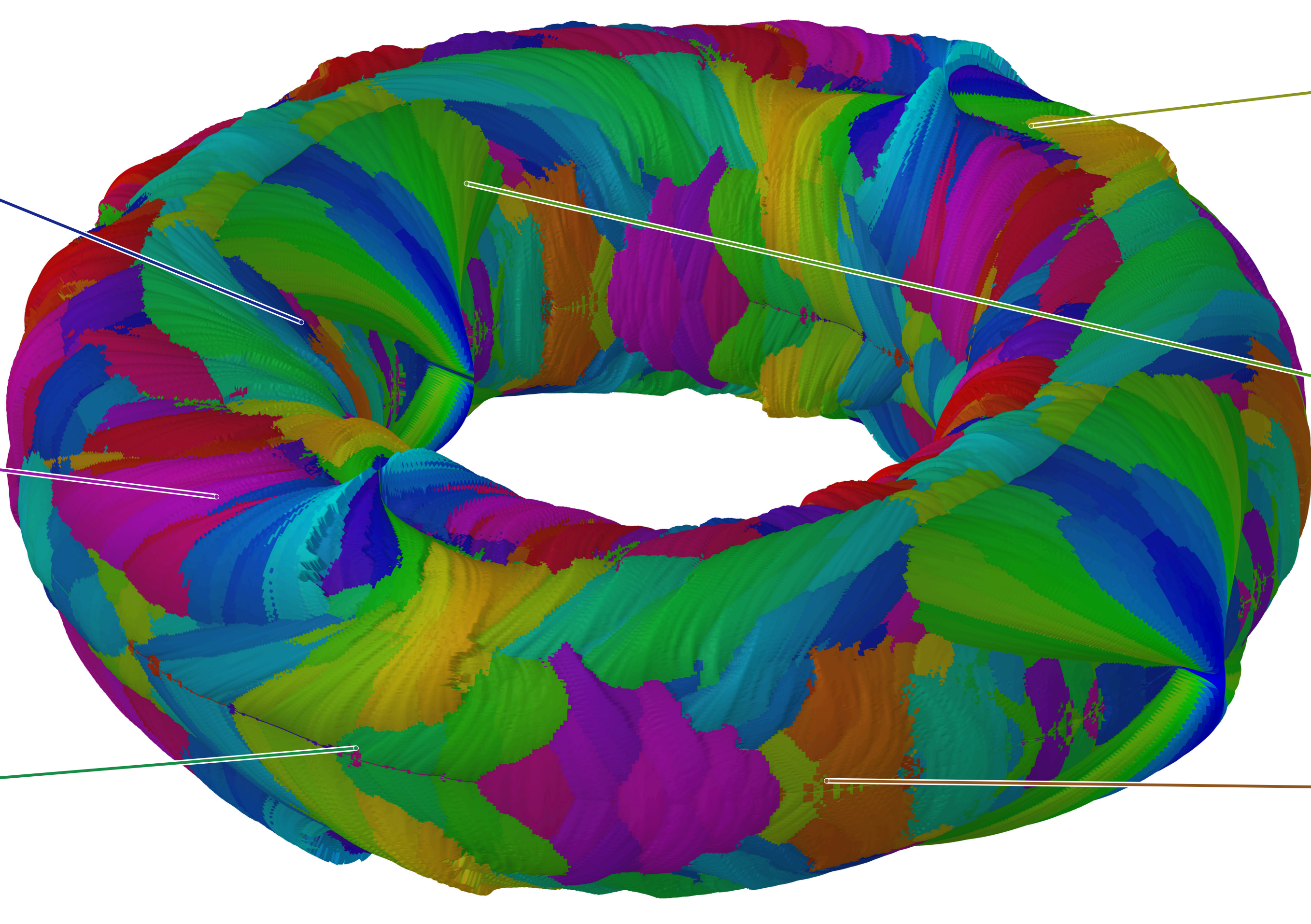
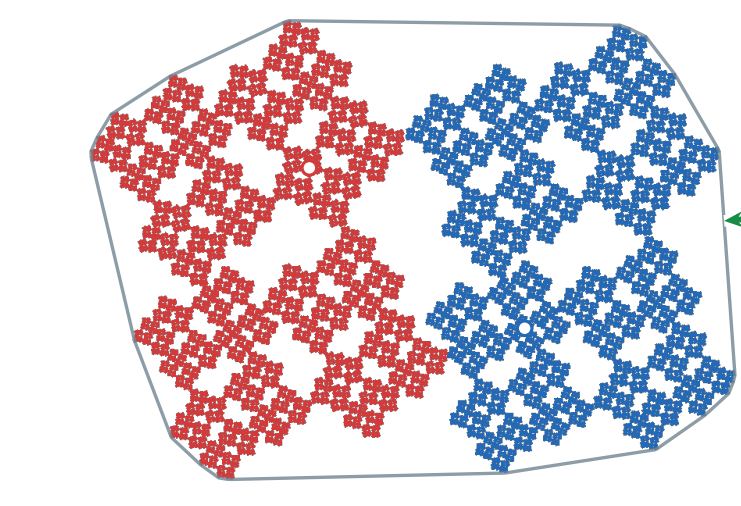
A $c_0 \simeq 1.376 + 0.512i$
 $c_1 \simeq 0.184 - 1.456i$



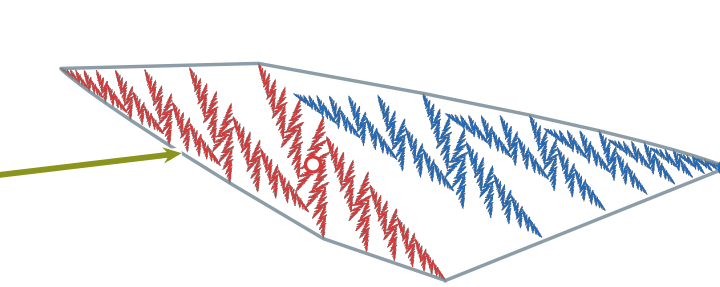
C $c_0 \simeq 1.167 + 1.143i$
 $c_1 \simeq -1.591 + 0.373i$



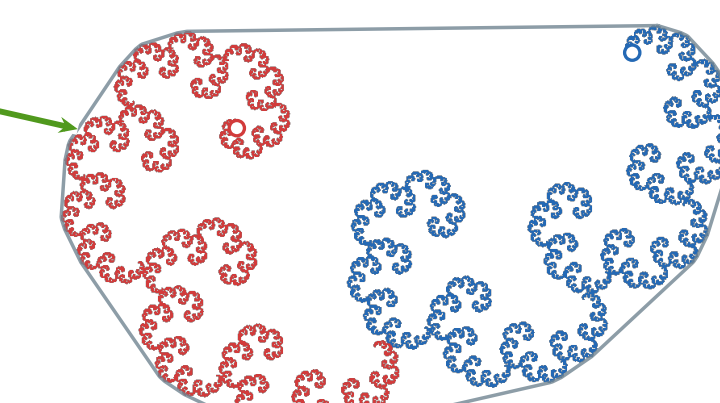
E $c_0 \simeq 0.307 + 1.442i$
 $c_1 \simeq 0.123 + 1.469i$



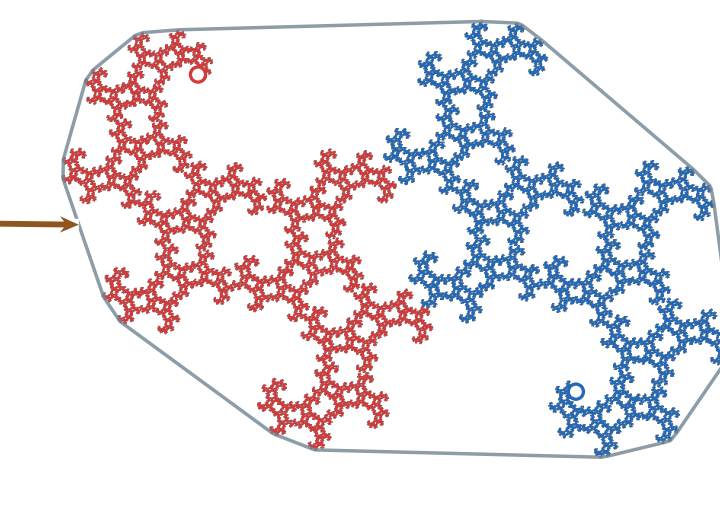
D $c_0 \simeq -1.437 - 0.534i$
 $c_1 \simeq 1.532 + 0.032i$



B $c_0 \simeq 0.572 + 1.445i$
 $c_1 \simeq 1.346 - 0.777i$



F $c_0 \simeq 1.247 + 0.866i$
 $c_1 \simeq 1.128 + 1.016i$

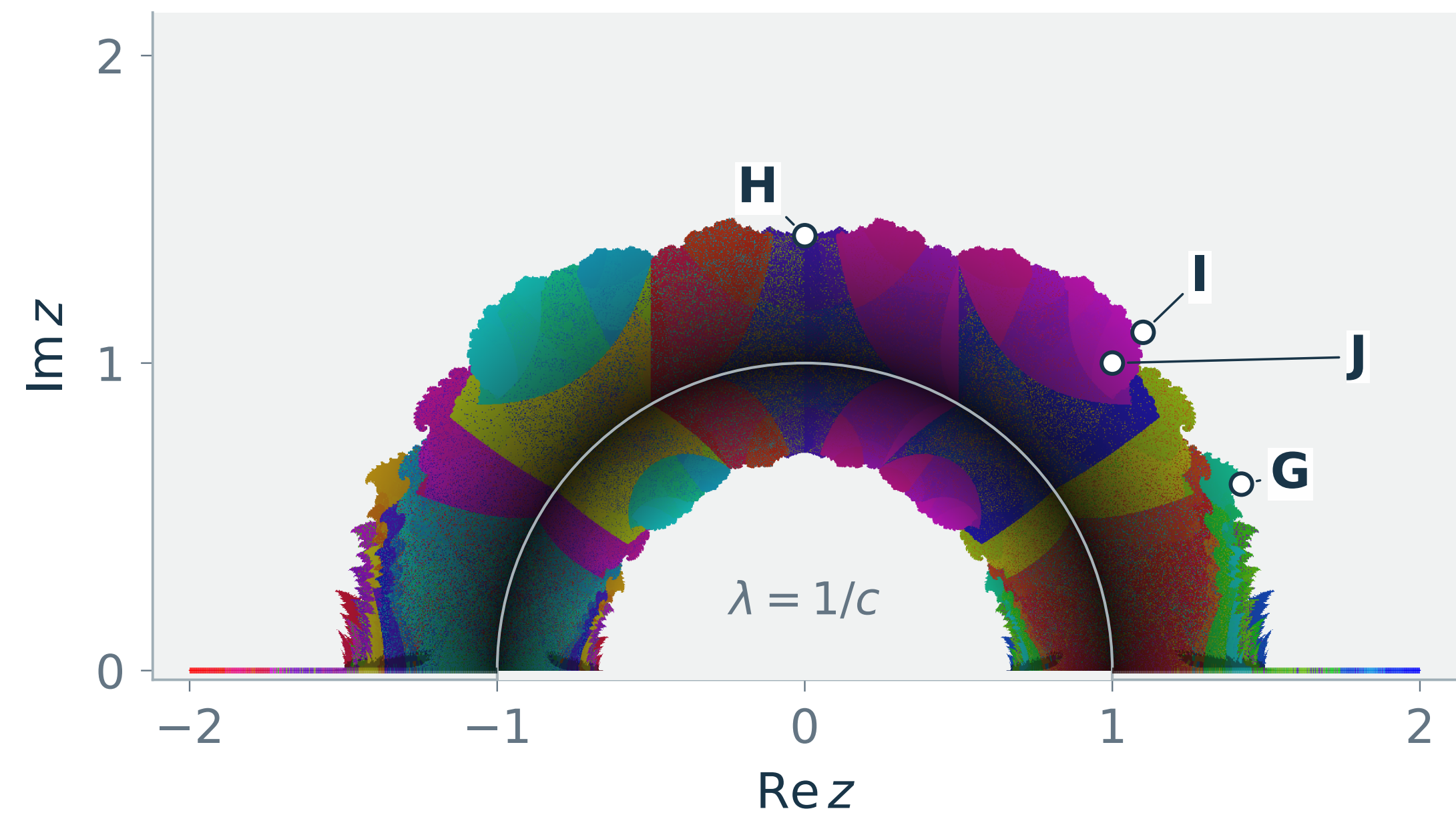


All three families: $r_0^2 + r_1^2 \geq 1 \Rightarrow$ connected [13]; $r_0 + r_1 < 1 \Rightarrow$ disconnected, $r_j = |c_j|^{-1}$.

2 Collinear slice

$c_0 = c_1 = c$ ■ binary fractals [1,5]

Explore this slice
complextrees.com/collinear



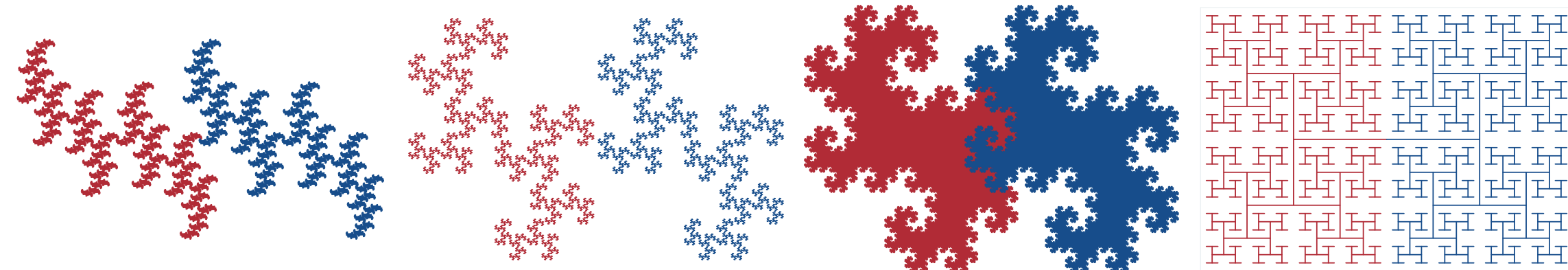
Connected and locally connected [5,11]; regular closed off $\mathbb{R}$ [9].

G Dendrite
 $c \simeq 1.420 + 0.606i$

I Dust
 $c = 1.1 + 1.1i$

J Twin dragon
 $c = 1 + i$

H Shared H-tree
 $c = i\sqrt{2}$

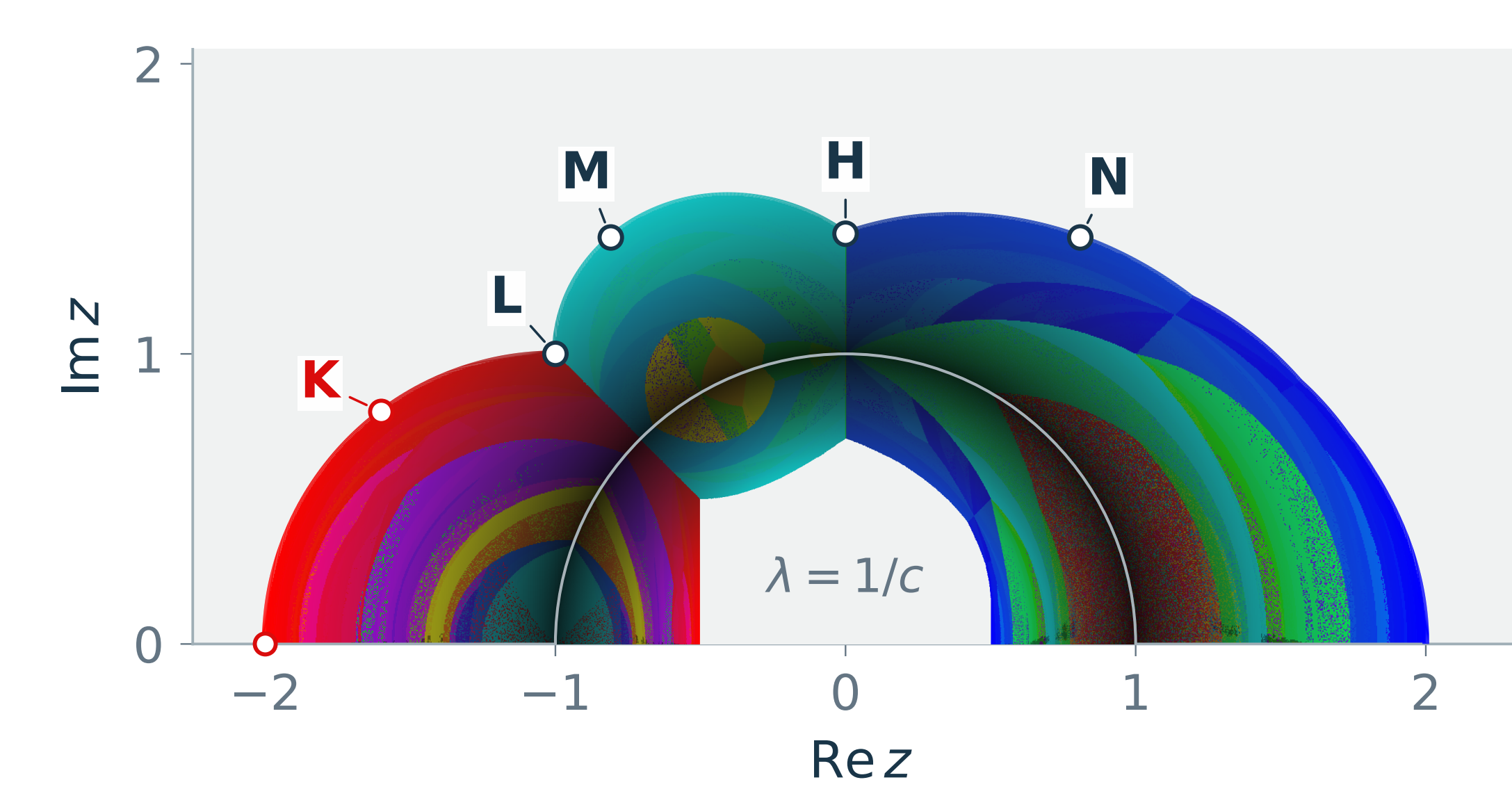


H: finite H-tree construction; its limiting attractor is a rectangle. $\varphi = (1 + \sqrt{5})/2$.
All slice charts: $z = c$ outside the unit circle; $z = \lambda = 1/c$ inside its upper half.

3 Mirror-symmetric slice

$(c_0, c_1) = (\bar{c}, c)$ ■ complex trees [2,3]

Explore this slice
complextrees.com/mirror



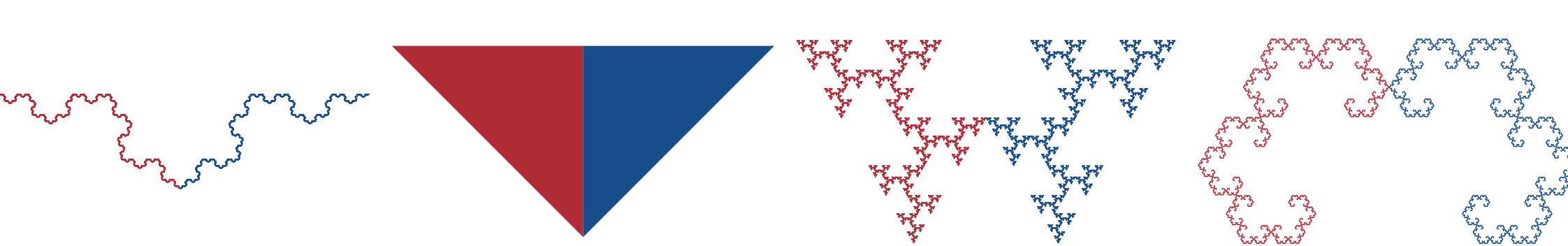
Connected, with a piecewise smooth boundary [10].

K Koch curve
 $c = -1.6 + 0.8i$

L Right triangle
 $c = -1 + i$

M 120° tree
 $c = \varphi e^{2\pi i/3}$

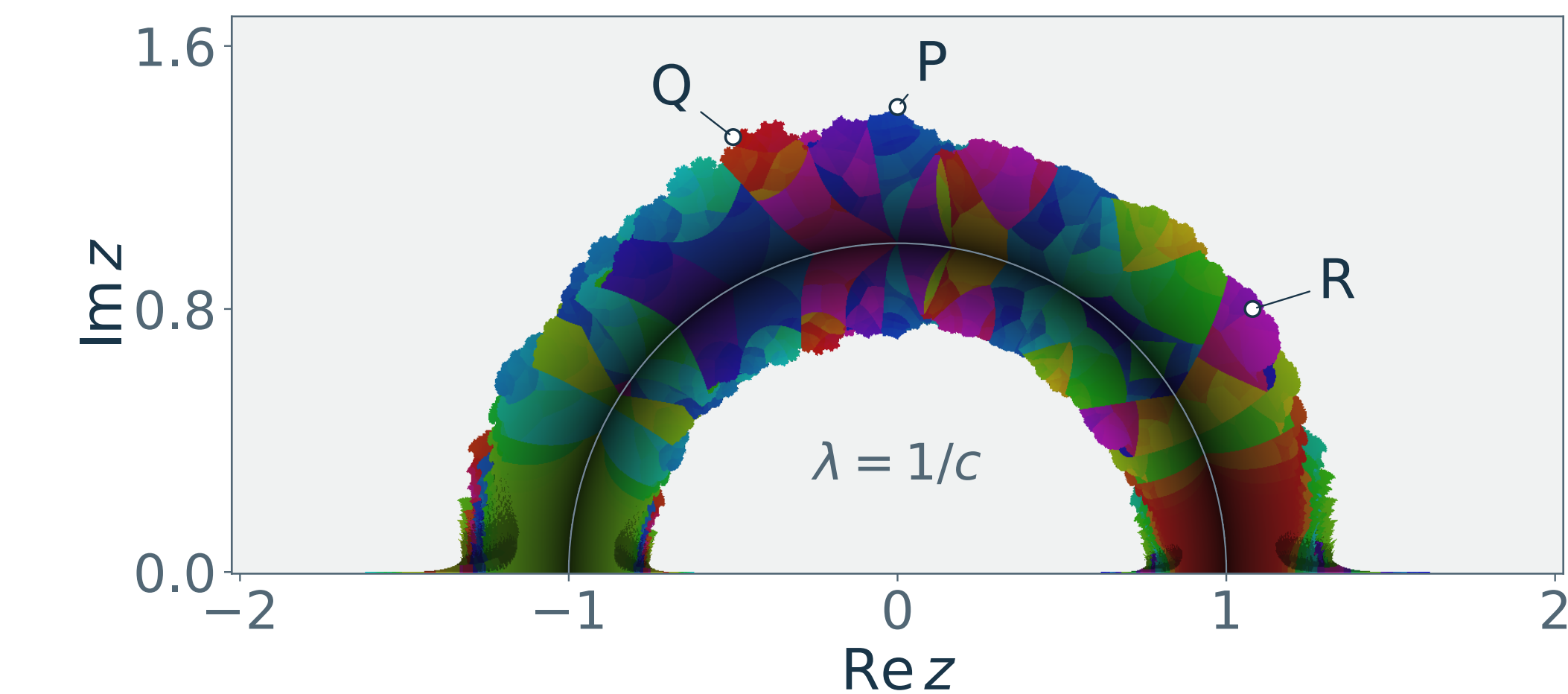
N 60° tree
 $c = \varphi e^{\pi i/3}$



4 Fibonacci–Mandelbrot

$(c_0, c_1) = (c, c^2)$ [6]

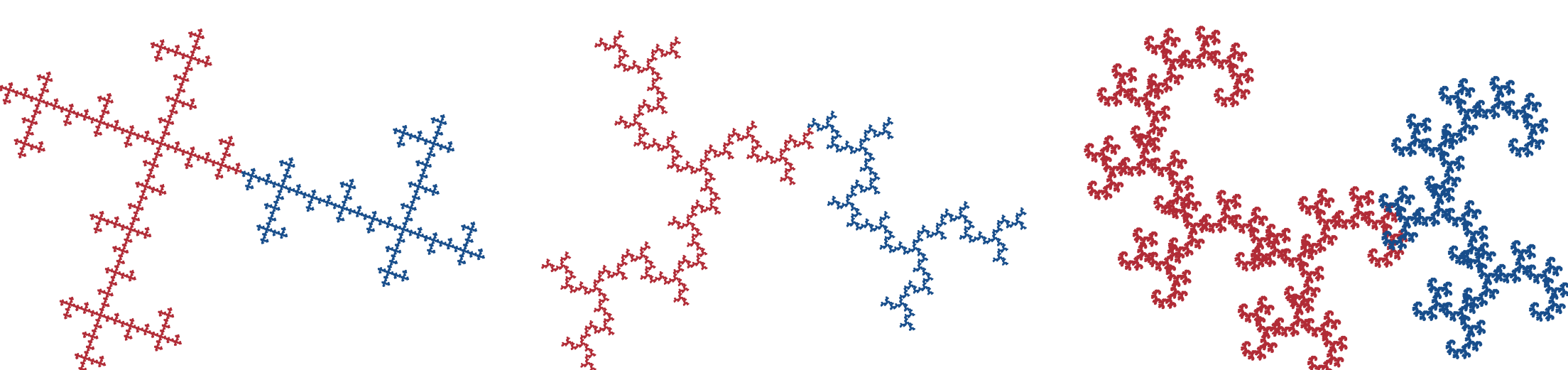
$w = |c|/(|c| + 1) > \frac{1}{2}$, with $|c| > 1$.



P $c = i\sqrt{2}$

Q $c = (-1 + i\sqrt{7})/2$

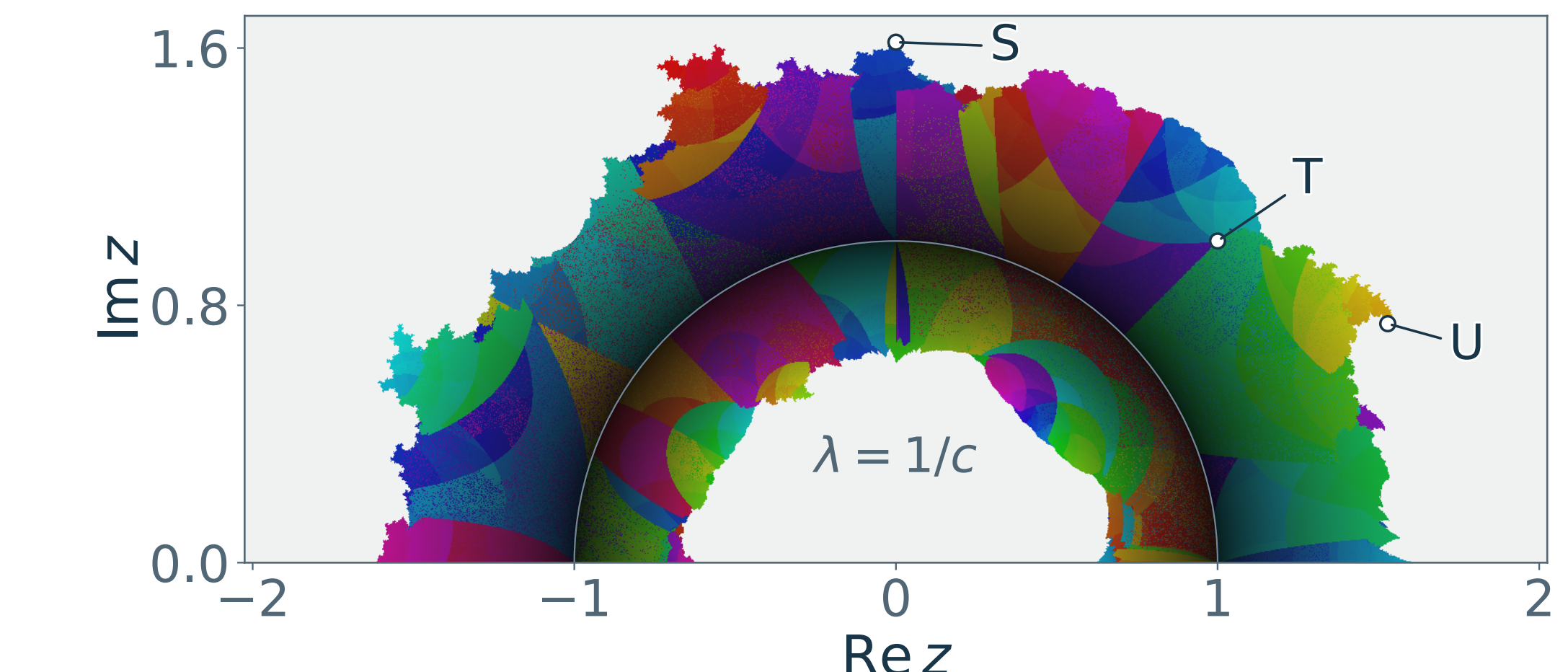
R $c = 1.08 + 0.80i$



5 Quarter-turn slice

$(c_0, c_1) = (c, ic)$ (90° relative rotation)

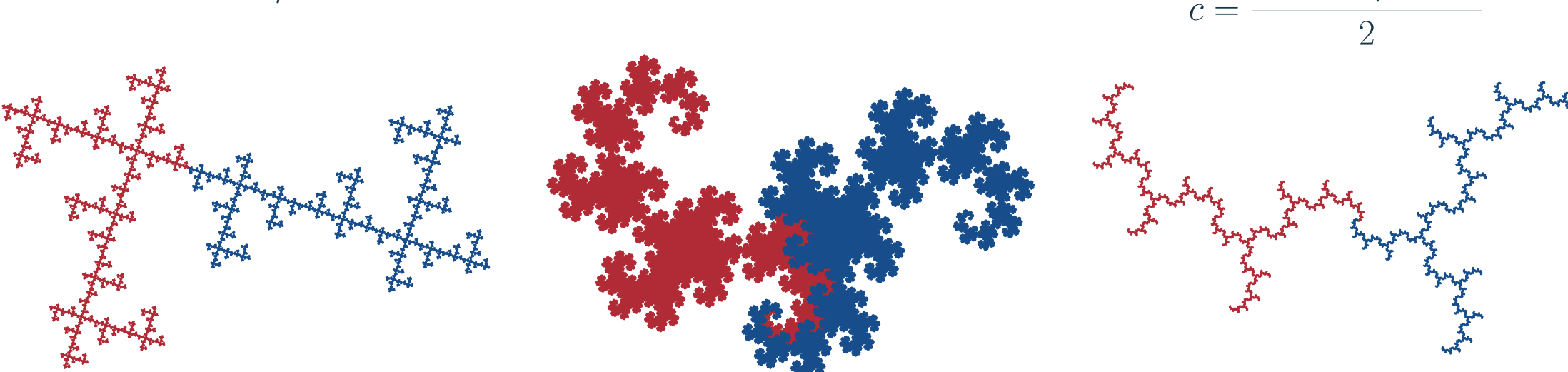
Equal contraction ratios: $w = \frac{1}{2}$, with $|c| > 1$.



S Golden-ratio boundary
 $c = i\varphi$

T Heighway dragon
 $c = 1 + i$

U Algebraic spiral tip
 $c = \frac{1 + i + \sqrt{4 + 2i}}{2}$



Fibonacci locus: connected and locally connected [6]. Quarter-turn slice: **S and U are exact boundary points**.
Finite-depth address colours; unresolved local hue estimates are additionally darkened.

Project, manuscript & interactive lab: complextrees.com/4D/

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